

Nonmonotone Rowley Templates and the Bound

$R_5(5) \geq 44,338$

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23 July 2026

Abstract

We give an explicit effective Rowley $(5, 5, 3)$ template of order 99 with extension parameter $\phi = 40$. Its inherited distance colours remain K_5 -free under period-98 repetition through the exact cutoff $4(99 - 2) = 388$, and its third distance class is sum-free. Combining it with an explicit linear $(5, 5, 5; 453)$ prototype by Rowley’s generalized construction yields a five-colouring of $K_{44,337}$ without a monochromatic K_5 . Consequently,

$$R_5(5) \geq 44,338.$$

We also certify that no effective $(5, 5, 3)$ template of order 98 has $\phi \geq 40$. Since templates with $\phi = 40$ exist at orders 97 and 99, feasibility at fixed extension threshold is nonmonotone in the order. The negative result is exhaustive: one DRAT certificate proves that the first forty distances have only twenty-two admissible two-colour prefixes, and eleven further DRAT certificates exclude every unrestricted tail up to global colour exchange. An independent semantic checker reconstructs all CNFs from explicit sum and five-vertex witnesses before proof checking. The positive objects are checked by independent Python and C++ clique searches, and the full compound word is reconstructed independently.

Keywords. Multicolour Ramsey numbers; linear colourings; triangle-free templates; SAT certificates; DRAT; nonmonotonicity.

MSC 2020. 05C55, 05D10, 68V15.

1 Introduction

For positive integers r and k , the diagonal multicolour Ramsey number $R_r(k)$ is the least N such that every r -edge-colouring of K_N contains a monochromatic K_k . Thus an explicit r -colouring of K_M without a monochromatic K_k proves $R_r(k) \geq M + 1$.

Rowley developed a composition method in which a linear colouring with a distinguished triangle-free template is combined with a second linear Ramsey colouring [1, 2]. His published effective $(5, 5, 3)$ template of order 93 and parameter $\phi = 40$, combined with an order-453 prototype, gives

$$R_5(5) \geq 41,626.$$

*Draft for independent review. Correctness, priority, and publication readiness are treated as separate questions.

Revision 18 of the standard survey records the same value [3].

This note provides an order-99 template at the same value of ϕ . The compound order is

$$(99 - 1)(453 - 1) + 1 + 40 = 44,337,$$

giving the asserted lower bound 44,338. The complete 44,336-symbol distance word is included in the accompanying artifact package.

The intervening failure at order 98 is not merely a failed search. It is proved by exhaustive case decomposition and checkable unsatisfiability certificates. Together with valid templates at orders 97 and 99, this proves that the set of feasible orders at fixed ϕ need not be an interval. The order-99 word uses a non-cyclic forty-symbol prefix, making symmetry breaking part of the mathematical mechanism rather than a search convenience.

The latest survey checked here is revision 18, dated 24 April 2026 [3]. Bounded searches found no competing order-99 template or the numerical bound 44,338. This is not a proof of priority; the claims of this note are correctness claims pending external current-literature review.

2 Linear templates

Definition 2.1 (Linear distance colouring). Let $m \geq 2$. A linear three-colouring of K_m is a word

$$c : \{1, \dots, m - 1\} \longrightarrow \{1, 2, 3\},$$

where edge $\{i, j\}$, $i < j$, receives colour $c(j - i)$. Put $p = m - 1$ and

$$A_1 = c^{-1}(1), \quad A_2 = c^{-1}(2), \quad T = c^{-1}(3).$$

The set T is the distinguished template colour. Its linear graph is triangle-free exactly when

$$\nexists a, b > 0 \quad a, b, a + b \in T, \tag{1}$$

including the case $a = b$. The extension parameter is

$$\phi = \min(T) - 1.$$

For $s \in \{1, 2\}$, repeat inherited colour s with period p by assigning positive distance d the base colour

$$c(((d - 1) \bmod p) + 1). \tag{2}$$

The representative of residue zero is therefore p , not 0.

Definition 2.2 (Effective (5, 5, 3) template). An order- m word is effective with parameter ϕ if:

1. $p = m - 1$ belongs to T and $\min(T) - 1 = \phi$;
2. T satisfies (1); and
3. for $s = 1, 2$, the periodically repeated colour in (2) contains no K_5 .

The infinite-looking final condition has an exact finite form.

Lemma 2.3 (Directed-Cayley cutoff). *Fix an inherited colour s and period p . Define a directed relation on $\mathbb{Z}_p$ by*

$$x \longrightarrow y \iff c(((y - x - 1) \bmod p) + 1) = s.$$

The periodic colour contains a K_5 if and only if there are distinct $r_0, \dots, r_4 \in \mathbb{Z}_p$ such that

$$r_i \longrightarrow r_j \quad (0 \leq i < j \leq 4).$$

If such a witness exists, one exists on integer vertices contained in $\{0, \dots, 4(p-1)\}$.

Proof. An increasing integer witness reduces modulo p to the displayed ordered relation. Equal residues cannot occur, because their difference is a multiple of p , represented by $p \in T$ rather than inherited colour s .

Conversely, for $i = 0, \dots, 3$, let $g_i \in \{1, \dots, p-1\}$ be the least positive representative of $r_{i+1} - r_i$. Put $x_0 = 0$ and $x_{i+1} = x_i + g_i$. Then $x_j - x_i$ has residue $r_j - r_i$ for every $i < j$, so the five x_i form a monochromatic clique. Moreover

$$x_4 = g_0 + g_1 + g_2 + g_3 \leq 4(p-1).$$

□

Lemma 2.3 is the $(5, 5, 3)$ specialization of the compression mechanism in Rowley's Theorem 3.2 [1]. It also identifies the native finite obstruction: an ordered transitive K_5 in a directed Cayley relation.

3 The order-99 certificate

Certificate 3.1 (Order-99 word). Concatenate the following two lines:

```
21112112111222112222222112221112112111232113211113323232333133331333
3113333133323332333111231123
```

The resulting word has length 98.

Proposition 3.2. *Certificate 3.1 is an effective $(5, 5, 3)$ template of order 99 with $\phi = 40$.*

Proof. Direct enumeration gives $98 \in T$, $\min(T) = 41$, $|T| = 33$, and no $a, b, a + b \in T$. Two independent exact clique searches form the repeated inherited-colour graphs on vertices $0, \dots, 388$ and reject K_5 in both colours. By Lemma 2.3, span $388 = 4(98 - 1)$ is complete. The Python implementation visits 1,645,997 and 917,897 search nodes for the two colours; the independent C++ implementation also returns VALID. □

The canonical file is `results/order99_linear_prefix8.template`, with SHA-256

```
2643001cc425898d584bd374e20928d1dbc6a72fd4011711151343d0ad072966.
```

Its first forty symbols contain twenty occurrences of each inherited colour and are not cyclically reflected.

4 The order-453 prototype

The second input is the 452-symbol word in Appendix A. It is recovered from the archived order-977 template in Rowley's ancillary data: the first 452 distances are the order-453 prototype and distance 453 begins the newly introduced template colour [2].

Proposition 4.1. *The word in Appendix A defines a cyclic linear three-colouring of K_{453} with no monochromatic K_5 .*

Proof. Independent Python bit-mask and C++ ordered-vector clique searches exhaustively reject K_5 in each colour. Their exact node counts are, respectively,

	1	2	3
Python	1,040,265	5,811,421	782,585
C++	237,855	837,235	201,199.

Direct comparison also gives $v(d) = v(453 - d)$ for $1 \leq d \leq 452$. □

The artifact `sources/rowley_exoo_order453.prototype` has SHA-256

19c97e6279c184f6f462786cadda4b7c9773d870a5b680a04eb1503ef384a2d0.

Once the word is explicit and Proposition 4.1 is checked, its historical recovery is no longer a logical dependency.

5 The compound colouring

Set $p = 98$. Let A_1, A_2, T be the classes of Certificate 3.1, and let B_1, B_2, B_3 be the prototype classes. For $s = 1, 2$, define

$$A'_s = \{\ell + (\mu - 1)p : \ell \in A_s, 1 \leq \mu \leq 452\} \cup \{\ell + 452p : \ell \in A_s, 1 \leq \ell \leq 40\}. \quad (3)$$

For $j = 1, 2, 3$, put

$$B'_j = \{\ell + (\mu - 1)p : \ell \in T, \mu \in B_j\}. \quad (4)$$

Lemma 5.1. *The five sets $A'_1, A'_2, B'_1, B'_2, B'_3$ partition $\{1, \dots, 44,336\}$.*

Proof. For $1 \leq d \leq 452p$, write uniquely $d = \ell + (\mu - 1)p$ with $1 \leq \ell \leq p$ and $1 \leq \mu \leq 452$. If $\ell \in A_s$, equation (3) assigns d to A'_s . If $\ell \in T$, exactly one B_j contains μ , and (4) assigns d to B'_j .

The remaining distances are $452p + \ell$ with $1 \leq \ell \leq 40$. Since $T \cap [1, 40] = \emptyset$, the second line of (3) assigns each uniquely. □

The five class sizes are

class	A'_1	A'_2	B'_1	B'_2	B'_3	total
size	16,292	13,128	4,488	6,402	4,026	44,336.

Theorem 5.2. *The distance classes in (3)–(4) define a five-colouring of $K_{44,337}$ without a monochromatic K_5 . Consequently,*

$$R_5(5) \geq 44,338.$$

Proof. Lemma 5.1 defines the colouring. We exclude the two types of output colour separately.

Inherited colours. Suppose output colour $s \in \{1, 2\}$ contains a K_5 . Translate its least vertex to zero and reduce all five vertices modulo p . Every ordered pairwise residue difference lies in A_s . Lemma 2.3 therefore produces a K_5 in the finite repeated graph on $0, \dots, 388$, contradicting Proposition 3.2.

Prototype-derived colours. Fix $j \in \{1, 2, 3\}$ and suppose output colour $j+2$ contains a K_5 . Translate its least vertex to zero. Each other vertex has a unique representation

$$d_i = t_i + (\mu_i - 1)p, \quad t_i \in T, \quad \mu_i \in B_j.$$

For $d_i < d_k$, membership of $d_k - d_i$ in B'_j forces $t_i \geq t_k$: if $t_i < t_k$, then the positive residue $t_k - t_i$ would belong to T , contradicting $t_i + (t_k - t_i) = t_k$ and sum-freeness.

Thus the residues t_i are nonincreasing along the ordered vertices. The prototype index of $d_k - d_i$ is then $\mu_k - \mu_i$, whether $t_i = t_k$ or $t_i > t_k$. Hence every positive difference $\mu_k - \mu_i$ lies in B_j . The indices are strictly increasing, so the five indices, including zero for the translated least vertex, form a monochromatic K_5 in prototype colour j , contradicting Proposition 4.1.

All five colours are K_5 -free. The order calculation gives

$$98 \cdot 452 + 41 = 44,337,$$

and the Ramsey inequality follows. □

The complete word is `results/r5_5_order44337.linear-coloring`, with SHA-256

274acbf17bf7732b16ef7d20c97486eb469486907fd1357c16990ed4332f7158.

It is linear and non-cyclic. A generator implementing the block formula and an independent literal set-union reconstruction agree on all 44,336 symbols.

6 Certified nonexistence at order 98

Theorem 6.1. *There is no effective Rowley $(5, 5, 3)$ template of order 98 with $\phi \geq 40$.*

Certificate architecture. Any candidate has period 97, terminal distance 97 in T , and $T \cap [1, 40] = \emptyset$. Thus the first forty distances form a two-colour linear colouring of K_{41} with neither colour containing K_5 .

Stage 1: exhaustive prefixes. For each five-set $V \subseteq \{0, \dots, 40\}$, let

$$\Delta(V) = \{|u - v| : u, v \in V, u \neq v\}.$$

If Boolean variable x_d selects inherited colour 1 at distance d , the two clauses

$$\bigvee_{d \in \Delta(V)} x_d, \quad \bigvee_{d \in \Delta(V)} \neg x_d$$

exclude a monochromatic K_5 in colours 2 and 1. Deduplication over all V gives 45,374 distinct distance sets and 90,748 clauses.

There are 22 satisfying prefix words, stored as 11 representatives up to global exchange of colours 1 and 2. Adding blocking clauses for those 22 words makes the prefix CNF unsatisfiable. The file `prefix41_exhaust.drat` is a verified DRAT proof of that UNSAT instance. Independently, the semantic checker regenerates all 45,374 sets, checks each representative directly, reconstructs the CNF byte-for-byte, and invokes the proof checker.

Stage 2: unrestricted tails. For each of the 11 representatives, introduce three one-hot variables for every distance $1, \dots, 97$. Fix only the selected first forty symbols and the terminal template colour. Every distance $41, \dots, 96$ is allowed all three colours.

For every $a, b > 0$ with $a + b \leq 97$, include the clause forbidding all of $a, b, a + b$ from lying in T . Each inherited-colour obstruction clause carries an explicit ordered witness

$$0 = v_0 < v_1 < \dots < v_4 \leq 384.$$

Its period-97 distance coordinates are

$$D(V) = \{((v_j - v_i - 1) \bmod 97) + 1 : 0 \leq i < j \leq 4\},$$

and the clause forbids all coordinates of $D(V)$ from receiving the corresponding inherited colour. Every such clause is therefore a valid necessary condition for a template.

All 11 tail CNFs are UNSAT, with verified DRAT proofs. The semantic checker validates every sum identity and every five-vertex witness, then reconstructs each CNF byte-for-byte before proof checking. Exchanging inherited colours covers the complementary 11 prefixes. Hence every possible first forty symbols and every unrestricted tail are eliminated. $\square$

Remark 6.2. The tail formulas need not contain every possible K_5 clause. Each recorded clause is valid, and UNSAT of a relaxation containing only recorded valid obstructions already proves that no candidate exists.

The certificate packet is `certificates/order98_phi40_exhaustion/`. It contains 12 CNF/DRAT pairs, semantic clause sidecars, prefix representatives, hash manifests, and the independent checker. The checker compiles against a pinned MIT-licensed copy of `drat-trim` [5].

Corollary 6.3 (Nonmonotonicity). *At fixed threshold $\phi = 40$, existence of an effective $(5, 5, 3)$ template is not monotone in the template order.*

Proof. The frozen word `results/order97_reflected.template` is accepted by both exact verifiers, Theorem 6.1 excludes order 98, and Proposition 3.2 supplies order 99. $\square$

7 Verification and reproducibility

All release-critical checks run with Python 3, a C compiler, a C++17 compiler, and a POSIX shell:

```
sh tests/run_certificate_checks.sh
```

The suite has no third-party Python dependency. It:

1. runs positive and deliberately corrupted fixtures against independent template verifiers;
2. checks all three prototype colours with two clique algorithms;
3. checks the order-97 and order-99 words with both algorithms;
4. regenerates the complete compound and independently reconstructs all output distance sets;
5. compiles the pinned `drat-trim.c`; and
6. semantically reconstructs all order-98 CNFs and verifies all 12 DRAT proofs.

GitHub Actions executes the same command on Ubuntu and macOS.

The two clique implementations share only the mathematical specification. The Python code uses arbitrary-precision bit masks; the C++ code uses ordered candidate vectors. Both exhaustively recurse on a candidate set intersected with the forward neighbours of each selected vertex. A branch is pruned only when fewer candidates remain than the number of required clique vertices.

The 44,337-vertex graph is not checked by a naive all-vertex clique enumeration. Its correctness follows from Theorem 5.2, whose two finite inputs are checked exactly.

8 Scope and literature status

Theorem 6.1 is exact only at order 98 and threshold $\phi \geq 40$. It says nothing about smaller ϕ or later orders. Solver reconnaissance at orders 100 through 105 produced no positive object, but those runs are not certificate-backed and are not claims.

Rowley’s 2022 paper gives $R_5(5) \geq 41,626$ [2]; revision 18 of the dynamic survey records that value [3]. The asymptotic update of Campos and Pohoata does not state a competing finite value [4]. Searches for 44,338, an order-99 effective template, and follow-ups to Rowley’s construction found no collision in the checked corpus.

Accordingly, the present status is:

finite correctness certified locally; external priority review required.

A Order-453 prototype word

Concatenate the following seven lines exactly:

```
1222333322121112131212231312221323223233222132322323322111112212111221
3132231212232313223121323231222132332111122233231221212233323221121321
2111223121122233232332332223323321111221111123131222213122313122
12132223121122233222112132223121221313221312222131321111221111233233
222332332332332332221121322111212312112232333221212213233222111123323
122213232312132231323221213223131221112122111122332322323122233232232
31222131322121312111212233332221
```

B Principal artifact hashes

Artifact	SHA-256
Order-99 template	2643001cc425898d584bd374e20928d1 dbc6a72fd4011711151343d0ad072966
Order-97 template	3287e4737fed6d355d59c4c08c3140a 829508039293914109c78000c7ba3a309
Order-453 prototype	19c97e6279c184f6f462786cadda4b7c 9773d870a5b680a04eb1503ef384a2d0
Complete compound	274acbf17bf7732b16ef7d20c97486eb 469486907fd1357c16990ed4332f7158
Prefix-exhaustion CNF	aec46291803dae6b29e180bde569d24 102fa5234aa3b8be096177b3eac1c179
Prefix-exhaustion DRAT	02ee6ff73c60629c7b9efbac2a95d32 6537d23c22e4e83a9380bb7c392aef57d

Acknowledgement

Computational exploration and technical drafting used automated coding and language-model assistance. Every mathematical claim in this note is tied to explicit source text, finite objects, or independently checkable certificates; no model output is used as authority.

References

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